

Math 237A HW 6

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1 D

Problem 1

Lazarsfeld Problem Set 5 (2):

Let $f : X \rightarrow Y$ be a dominant morphism of irreducible (quasi-projective) varieties. Prove that $\dim Y \leq \dim X$.

Solution: Given that $f : X \rightarrow Y$ is a dominant morphism, then the induced ring homomorphism $f^* : k[Y] \rightarrow k[X]$ is injective. Which, such ring homomorphism can be extended to the field homomorphism between their fraction fields, say $\overline{f^*} : k(Y) \rightarrow k(X)$ (or between their field of rational functions).

Now, suppose $\dim Y = n$, then there exists an algebraically independent subset over k with length n , say $g_1, \dots, g_n \in k(Y)$. So, the field homomorphism $\overline{f^*}$ must have $\overline{f^*}(g_1), \dots, \overline{f^*}(g_n) \in k(X)$ be algebraically independent also:

Suppose the contrary that they're algebraically dependent, then there exists $p(x_1, \dots, x_n) \in k[x_1, \dots, x_n]$, such that $p(\overline{f^*}(g_1), \dots, \overline{f^*}(g_n)) = 0$. Then, since $\overline{f^*}$ is a field homomorphism, one has $0 = p(\overline{f^*}(g_1), \dots, \overline{f^*}(g_n)) = \overline{f^*}(p(g_1, \dots, g_n))$, showing $p(g_1, \dots, g_n) = 0$ by the injectivity of field homomorphism. So, $g_1, \dots, g_n$ is no longer algebraically independent over k , which is a contradiction.

So, $\overline{f^*}(g_1), \dots, \overline{f^*}(g_n) \in k(X)$ is algebraically independent over k , showing $n \leq \text{trdeg}_k k(X) = \dim X$. Hence, $\dim Y \leq \dim X$.

2 ND

Problem 2

Lazarsfeld Problem Set 5 (3):

Given $d \geq 2$, denote by $\mathbb{P}^{N(d)}$ the projective space parametrizing all plane curves of degree d . (So $N(d) = \binom{d+2}{2} - 1$).

(a) Show that if $d = 2$, then there is a dense open subset $U \subset \mathbb{P}^5$ such that all conics corresponding to points in U are projectively equivalent, i.e. differ by a linear change of coordinates.

(b) On the other hand, prove that the analogous statement fails when $d \geq 3$.

Note: You may grant that $\text{SL}(n+1)$ is irreducible. You will want to observe that it has dimension $= (n+1)^2 - 1$.

3 ND

Problem 3

Lazarsfeld Problem Set 5 (4):

Find the dimension of the space $M_{n \times m}^{\leq r}$ of all $n \times m$ matrices of rank $\leq r$.

4 ND

Problem 4

Hartshorne 1.1.10:

- (a) If Y is any subset of a topological space X , then $\dim Y \leq \dim X$.
- (b) If X is a topological space which is covered by a family of open subsets $\{U_i\}$, then $\dim X = \sup \dim U_i$.
- (c) Give an example of a topological space X and a dense open subset U with $\dim U < \dim X$.
- (d) If Y is a closed subset of an irreducible finite-dimensional topological space X , and if $\dim Y = \dim X$, then $Y = X$.
- (e) Give an example of a Noetherian topological space of infinite dimension.